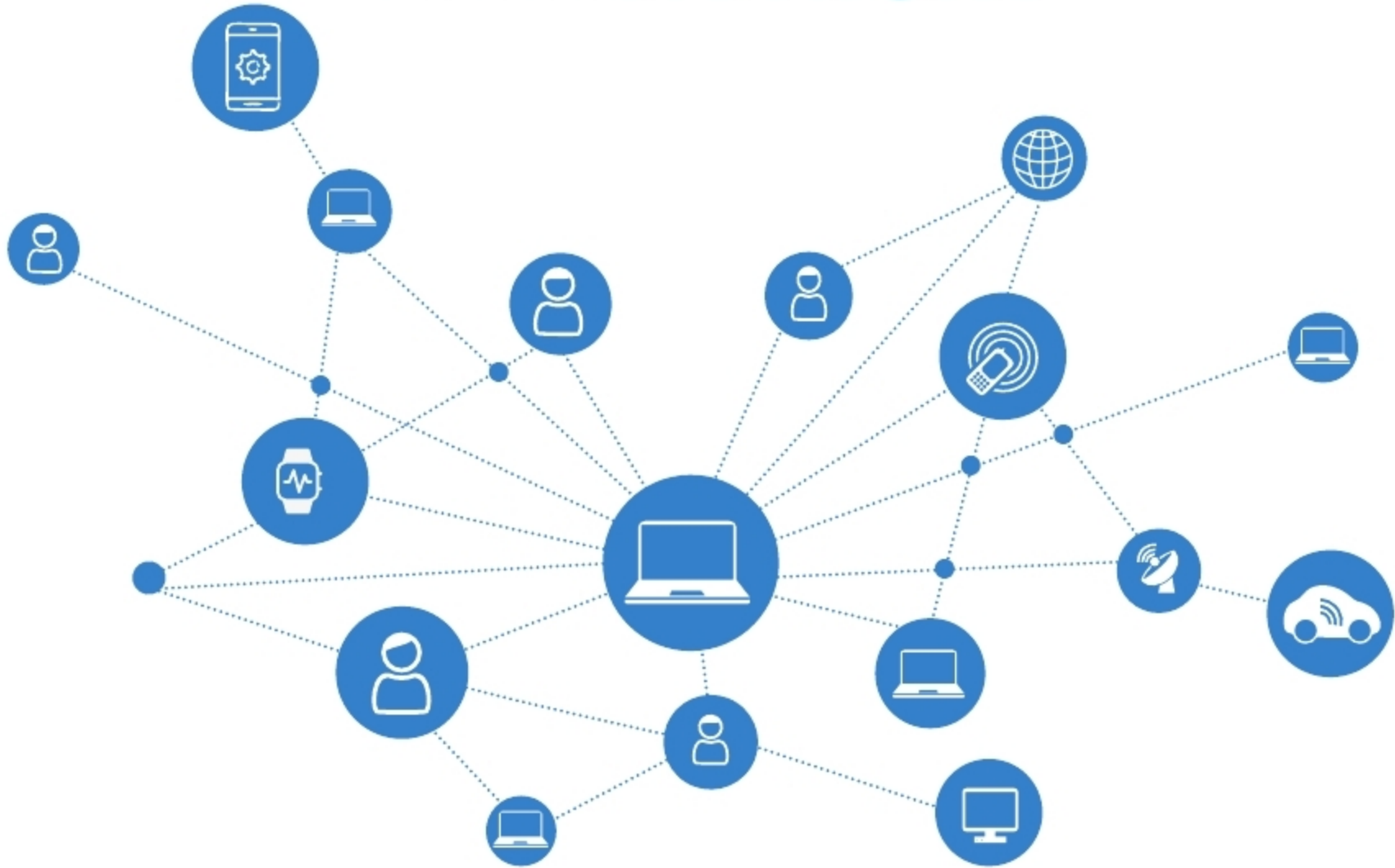


Gaining a Better Understanding of our Network



A network connects computers, mobile phones, peripherals and IoT devices, providing communication between them. It also facilitates communication between different networks and even with the Internet. So you must try and get to know it better.

For better understanding of the configuration of the computer network surrounding you, you should start from the machine that you are currently logged in to. This will give you all the necessary network configuration parameters, like the IP address of the present host, the DNS configuration, and which other machines are connected to and can communicate with your network.

Finding configuration information

In a UNIX platform, the easiest way to identify the basic configuration of a host is to use *hostname*, *ifconfig* and *netstat* commands. A few system text files are also useful to find out details about the internal configuration. These include */etc/nsswitch.conf*, */etc/resolv.conf*, and so on. Using this information, you can directly determine the identity of your machine and its location.

A little study of the *ifconfig* command can provide you with lots of basic network configuration and communication information of the connected network devices. For example,

the following listing of the *ifconfig -a* command gives the Internet address and *netmask* of the connected network device and can explore the underlying network configuration. If the host machine has multiple network interfaces, then for each device it will show the corresponding device network information.

```
>ifconfig -a
eno16777984: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>mtu 1500
inet 172.16.4.102 netmask 255.255.252.0 broadcast
172.16.7.255
        inet6 fe80::20c:29ff:fe3d:4db0prefixlen 64 scopeid
0x20<link>
ether 00:0c:29:3d:4d:b0 txqueuelen 1000 (Ethernet)
RX packets 1869057322 bytes 295709999938 (275.4 GiB)
RX errors 0 dropped 4028609 overruns 0 frame 0
TX packets 2499818214 bytes 798685297206 (743.8 GiB)
TX errors 0 dropped 0 overruns 0 carrier 0
collisions 0
```